Support, Resistance, and Technical Trading

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Abstract

I use coarse Bayesian updating to explain asset price levels at which the price will tend to stop and reverse. With limited attention, traders discretize the space of expected prices. I show that it is ex ante optimal for them to select an equally-spaced price grid to minimize ex post trading errors. Support and resistance levels appear in equilibrium prices because, as prices near a support (resistance) level, coarse Bayesians believe that the asset is underpriced (overpriced), leading to purchases (sales) and ensuing upward (downward) pressure on prices. I show that this phenomenon is robust to the presence of rational arbitrageurs.

JEL Classification: G14; G41; D84; D91.

Keywords: Technical trading; Behavioral finance; Coarse Bayesian updating; Price barriers

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1 Introduction

In asset markets, a support (resistance) level is a price level at which a decreasing (increasing) price will tend to change direction and begin to increase (decrease). This type of language is commonplace in Wall Street newsletters. For example, a quote from a *Market Watch* article written by Ashbaugh (2021) says:

> The February peak (3,950) remains an inflection point, and is followed by the 3,915 support.

In fact, according to Brock et al. (1992), trading range break—a strategy involving buying (selling) when prices penetrate a resistance (support) level—is one of the two most commonly used technical trading strategies. These authors famously find that such strategies are, indeed, effective at predicting stock price changes. Support and resistance levels also go by other names. For example, Shiller (2015) describes quantitative anchors, or "levels of the market that some people use as indications of whether the market is over- or underpriced and whether it is a good time to buy." To make ideas concrete, he mentions the nearest milestone of a prominent index such as the Dow or the nearest round number. As a final example, Donaldson and Kim (1993) find that the Dow Jones Industrial Average not only deflects off of price increments of one hundred, but also, having broken through an increment, the index then moves more than otherwise warranted.[1] The authors call such deflection points "price barriers."

In this paper, I develop a theoretical model explaining why support and resistance levels may be equilibrium outcomes of trading strategies when attention is limited. To model limited attention, I combine two existing theories. The first is coarse Bayesian updating from Jakobsen (2021): traders must select the most likely posterior from an exogenous set of posteriors.[2] Second, to make the set of feasible posteriors endogenous, I assume that traders' posterior means can only take on a fixed number of values, similar to the setup of Gul et al. (2017). Then when traders face Uniformly distributed prices,

[1] See also Koedijk and Stork (1994). In a theoretical framework, Balduzzi et al. (1997) investigate how the presence of buy-and-hold investors, who exogenously trade when the asset's price reaches a price barrier, affects equilibrium prices.

[2] This is related to categorical thinking from Mullainathan (2002) and hypothesis testing from Ortoleva (2012).

a natural result obtains: attention-constrained traders choose an *equally*-spaced price grid in order to minimize expected trading errors. Interestingly, Uniform risk (in terms of the ergodic distribution of prices) is an equilibrium object in the sense that, absent any coarse Bayesians, the otherwise-standard martingale economy generates a Uniformly distributed ergodic price distribution.

Consider the resulting trading strategies in the following simple example. Say that traders can only form expectations over two values: a low or a high price. Locally, demand functions are downward-sloping for the usual reasons. However, if higher prices today are an indication of higher prices tomorrow, a price increase may cause the trader's expectation to jump from the low to high value. In anticipation of my results, call this jump point the price barrier. First, consider when the price is below the price barrier. As the price approaches the barrier from below, the trader believes the asset is increasingly overpriced (relative to the low price expectation) and sells. This puts downward pressure on price, and hence the price tends not to cross the barrier. Symmetrically, consider when the price is above the barrier. As the price approaches the barrier from above, the trader believes the asset is increasingly underpriced (relative to the high price expectation) and buys, putting upward pressure on price. Again, the price tends not to cross the barrier.

I establish an equilibrium in two steps. Traders begin with an equilibrium conjecture about the distribution of future prices. Market orders are submitted based on traders' coarse beliefs, and prices then adjust based on the relative buy versus sell pressure (which includes noise trading alongside aforementioned coarse trading). I require the resulting distribution of prices to match that of the original conjecture. However, this "partial equilibrium" takes the trader's ex ante attention choice as given. So in addition, the equilibrium requires the following. Before trading, traders use the ergodic distribution of prices implied by the economy to optimally partition the space of price expectations, subject to an exogenous constraint on the number of partition cells.

I verify several properties of this equilibrium characterized by i) support and resistance levels and ii) equally-spaced posterior means. First, I verify that the formal definition of support and resistance from the empirical literature is satisfied in my model. In particular, Donaldson and Kim (1993) require that prices are a sub- (super-) martingale at the bottom (top) of each price range $p \in [\varepsilon i, \varepsilon(i+1)]$ where $\varepsilon = 100$ in Donaldson and

Kim (1993) and i denote integers. Consider how this requirement connects to properties of my equilibrium: i) coarse Bayesians buy (sell) at the bottom (top) of any given price range, and ii) this pattern repeats in other price ranges due to posterior means being equally-spaced. Second, I add rational traders to my economy previously populated only by coarse Bayesians (and noise traders) to show that rational arbitrage exacerbates, not alleviates, the support and resistance problem.

Related Literature

Following Brock et al. (1992), the academic debate pertaining to the usefulness of technical analysis has continued. Papers like Lo et al. (2000) find that several technical indicators provide incremental information and have practical value, while papers like Bajgrowicz and Scaillet (2012) call into question the economic value of technical trading altogether.[3] The benchmark for many of these papers is to show that technical trading beats buy-and-hold in terms of the Sharpe ratio, while accounting for transaction costs. Given i) the large annual growth rate of the average U.S. stock and ii) the difference in transaction costs between technical trading and buy-and-hold, this benchmark is quite high. This gives my theory—which simply connects support and resistance levels to limited attention—ample room to operate.

The use of coarse reasoning in financial markets is not new. Eyster and Piccione (2013) and Steiner and Stewart (2015) assume that the world evolves according to a Markov process, and model incomplete theories of the world using partitions of the true state space. Unlike my setting, both papers model risk neutral traders in the style of Harrison and Kreps (1978) and assume that traders are correct, on average, within each cell of the partition.

Gul et al. (2017) also partition their state space, and do so ex ante optimally given the constraint that a partition can only have k cells. There are several similarities between our papers: for example, the price space gets optimally partitioned into *intervals*. The first key difference between our papers is that Gul et al. (2017) consider a standard consumption problem, aside from consumers being restricted to consumption plans that

[3] Others in this debate include Zapranis and Tsinaslanidis (2012), Garzarelli et al. (2014), Zhu et al. (2015), and Marshall et al. (2009).

are measurable with respect to their partition. In my paper, I consider a speculative setting where only price expectations are coarse; trading choices need not be (because these may depend on the current price). Second, our set of findings is quite different: while they focus on price extremity, I focus on support and resistance levels.

A related paper is Gabaix (2014) in the sense that an ex ante attention allocation problem precedes an ex post consumer problem. However, the attention cost function studied there leads to sparsity, meaning that the agent chooses to pay attention to only a few important variables. While not directly related to financial markets, Matejka (2016) derives discrete pricing as a result of a monopolist's problem with standard rational inattention constraints.

My paper is also tangentially related to a literature, reviewed in Hommes (2001), that uses nonlinear dynamics to model financial markets. The goals of this literature are not so different from those of this paper: to explain technical trading and match realistic price moments. However, the mechanisms at work are very different. Most of these papers use adaptive belief systems, where fundamentalists trade based on news, chartists trade based on price extrapolation, and the choice of strategy (fundamentalist versus chartist) is based on an evolutionary fitness measure. Based on evolutionary forces, these papers show that there can be switching between phases of minor fluctuations, phases of optimism, and phases of pessimism. Most closely related to this paper is an exchange rate model of Westerhoff (2003), who augments the chartist-fundamentalist setup with traders who round to the closest whole number.

The remainder of this paper is organized as follows. In Section 2, I introduce coarse Bayesian updating, describe the model, and define my equilibrium concept. In Section 3, I solve the ex ante attention problem and investigate other properties of the equilibrium. In Section 4, I verify that the support and resistance definition is satisfied and discuss rational arbitrage. Section 5 concludes.

2 Model

2.1 Martingale Economy

In this section, I describe i) noise trading and ii) the process by which prices are formed. Together, these two elements form a baseline setting, to which coarse Bayesians will be appended later on. Time is discrete, infinite, and indexed by t, and there is a mass of noise traders who trade an amount u_t per period; u_t is drawn independently across periods from a density $h(u)$. I assume that this density $h(u)$ is symmetric around the point zero, where positive (negative) values are understood as long (short) positions. Furthermore, I assume that $\text{Var}(u_t) < \infty$ and that $\sup_{u \in \mathbb{R}} h(u) < \infty$.[4]

Prices of a single, durable asset paying no dividends are denoted p_t, and traders submit price-inelastic market orders. Given that orders are price-inelastic, market clearing is not possible and, instead, there is a market maker (or liquidity provider) who takes the opposite side of excess demand to clear markets. There are two standard market maker models in the literature: that of adverse selection like Kyle (1985) and that of inventory like Stoll (1978). With no adverse selection concerns in my model, I follow the inventory paradigm. While details of inventory models may differ in the sense of market maker risk aversion or capital constraints, all models have the same main prediction: excess demand (supply) puts upward (downward) pressure on ensuing prices.

Consistent with this discussion, I focus on linear equilibria where returns are linear (and increasing) in excess demand:

$$p_{t+1} = p_t + c \cdot \underbrace{u_t}_{\text{Excess demand}} \tag{1}$$

The market maker maps excess demand into higher prices and excess supply into lower prices at elasticity $c > 0$, which is an endogenous object but will be easily determined, as I show below. Equation (1) is consistent with empirical evidence from Chordia et al. (2002), who confirm that excess buy (sell) orders drive up (down) returns, even at lagged time periods.

[4] That the density is bounded is a technical condition and only used in Lemma 1.

To see how (1) fits with the inventory setting, consider the revenues and costs of the market maker. In terms of costs, the market maker loses money based on the position she must take, which is $-u_t$ to effectively clears markets. To see why she loses money, I take this position multiplied by returns, so that inventory returns for the market maker are

$$\mathcal{IR}_t \equiv -(p_{t+1} - p_t)u_t = -cu_t^2 \leq 0$$

where the equality follows from (1). In terms of revenues, market makers make revenues from transaction costs, which I assume are present in the form of a quadratic transaction cost levied on all traders. More precisely, these costs (and, hence, revenues for the market maker) are $\frac{\tau}{2}u_t^2$, where $\tau > 0$ is taken as an exogenous parameter of the model.[5] Then, for this market maker to make zero expected profits, I can simply set $c = \tau/2$.[6] These revenues and losses described above are consistent with evidence from Sofianos (1995), that market makers incur positioning losses on their inventory, which are compensated by revenues from transaction costs. For a microfounded market maker with the same qualitative properties as the one described here, see Teeple (2023).

2.2 Ex Post Problem

In this section, I append a mass of identical, long-lived, myopic coarse Bayesians to the previous economy. Preferences are mean-variance with quadratic transaction costs:[7]

$$u(p_t, x_t) = (\tilde{\mathbb{E}}[p_{t+1}|p_t] - p_t)x_t - \frac{\tau}{2}x_t^2 - \frac{\rho}{2}\tilde{\mathrm{Var}}(p_{t+1} - p_t|p_t)x_t^2,$$

where holdings of the asset are given by x_t, the risk aversion coefficient is given by $\rho > 0$, and the transaction cost parameter from Section 2.1 is given by $\tau > 0$. The optimal

[5] Temporary (permanent) price impact maps to the parameter τ (c).

[6] This continues to be true even when an additional mass of traders with demand $x(p_t)$ is appended to the model:

$$\underbrace{\mathbb{E}[c\{u_t + x(p_t)\}^2]}_{Losses} = \underbrace{\mathbb{E}\left[\frac{\tau}{2}u_t^2 + \frac{\tau}{2}x(p_t)^2\right]}_{Revenues} \implies c = \tau/2$$

[7] In the background, there is a zero-interest riskless asset, against which the risky asset is traded.

demand is:

$$x(p_t) = \frac{\tilde{\mathbb{E}}[p_{t+1}|p_t] - p_t}{\rho \tilde{\mathrm{Var}}(p_{t+1} - p_t|p_t) + \tau} \tag{2}$$

Importantly, price moments are denoted with a tilde because I assume that traders form expectations using *coarse Bayesian updating*. Jakobsen (2021) defines coarse Bayesian updating by partitioning the set of posteriors and imposing three properties:

1. Cells of the partition are convex.
2. The feasible belief (for each cell) lies within that cell.
3. If the Bayesian update falls within a cell, the corresponding feasible belief becomes the posterior.

My updating rule will be a special case of coarse Bayesian updating. To guarantee that the three properties hold, I first fix feasible beliefs $\mathcal{P}$ then define the partition using the following distance:[8]

$$\tilde{\mathbb{E}}[p_{t+1}|p_t] \in \mathrm{argmin}_p |p - \mathbb{E}[p_{t+1}|p_t]| \text{ s.t. } p \in \mathcal{P} \tag{3}$$

The distance function I employ in (3) is the distance between the *mean* of the true posterior and that of the feasible belief. Why focus on the mean when traders care about the first two price moments? My setting will have the following property: the true price variance, $\mathrm{Var}(p_{t+1} - p_t|p_t)$, is constant in p_t. Therefore, there is no updating in the variance, the coarse variance equals the true variance, $\tilde{\mathrm{Var}}(p_{t+1} - p_t|p_t) = \mathrm{Var}(p_{t+1} - p_t|p_t)$, and the only errors incurred from coarseness are through the mean in (3).

To get a sense of what the demand function (2) subject to (3) looks like, I fix three equilibrium objects: the first two are $\mathbb{E}[p_{t+1}|p_t] = p_t$ and $\mathrm{Var}(p_{t+1} - p_t|p_t) = 1$. Because both moments lack tildes, they refer to the true—not perceived—values. Absent any coarseness, these equilibrium values imply no trade. Finally, for the purpose of this example, I fix feasible posterior means $\mathcal{P}$ in (3) to equal the set $\{i\varepsilon + \varepsilon/2\}_{i \in \mathbb{Z}}$, which are equally-spaced posterior means with spacing ε. The interpretation is that traders form

[8] That the argmin in (3) need not be unique plays a minor role in simulations because prices leading to non-uniqueness have measure zero, and in theoretical results because integrals are taken.

expectations using rules of thumb and incur rounding errors. Because such mistakes do not map well to institutional traders or algorithms, these traders should be thought of as retail investors. Finally, I fix $\varepsilon = \rho = 1$ and $\tau = 0.1$. The resulting demand function is plotted in Figure 1.

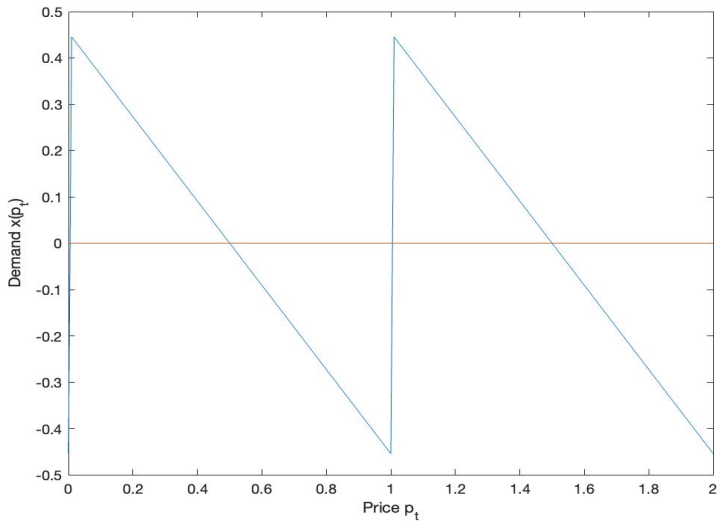

Figure 1: Coarse Demand Function

Demand curves slope downwards for the usual reasons. However, at certain prices, traders use higher prices in period t as a signal for higher prices in period $(t+1)$ and shift their demand upwards. The fact that traders use prices in t to predict prices in $(t+1)$ follows from $\mathbb{E}[p_{t+1}|p_t] = p_t$. And demand shifts discretely because expectations (the demand intercept) shift discretely. While my model only allows for market orders, if the entire demand schedule were to be submitted, locally downward sloping intervals would correspond to standard "limit orders," while upward discontinuities in demand would correspond to "stop orders."

As noted by Mullainathan (2002), coarse Bayesians can both under- and over-react to news. Here, the news is simply the current price realization. As a point of reference, say that $p_0 = 0.5$ in Figure 1, so the posterior mean is $\tilde{\mathbb{E}}[p_1|p_0] = 0.5$. For modest price increases, for example $p_1 = 0.9$, the posterior mean remains unchanged, $\tilde{\mathbb{E}}[p_2|p_1] = 0.5$; this represents an under-reaction with respect to a standard Bayesian. For larger price

increases, for example $\hat{p}_1 = 1.1$, the posterior mean moves more than that of a Bayesian, $\tilde{\mathbb{E}}[p_2|\hat{p}_1] = 1.5$; this represents an over-reaction.

2.3 Ex Ante Problem

In this section, I introduce the coarse Bayesian trader's attention problem. Similar to the setup of Gul et al. (2017), I assume that posteriors can be ex ante optimally chosen subject to the constraint that there can be no more than k posteriors (per fixed interval of prices). This restriction should be interpreted as resulting from limited attention.[9] Unlike the per-period trading problem, I assume that this attention problem is solved once at time $t = 0$.

One complication in my setup, not present in Jakobsen (2021) or Gul et al. (2017), is that prices are unbounded. In particular, I allow the space of prices to be all of $\mathbb{R}$ because the only variable of interest in my setting is the price *difference*, $p_{t+1} - p_t$.[10] To work around this issue, I assume that each interval of prices, $\{..., [-2b, -b], [-b, 0], [0, b], [b, 2b], ...\}$ can contain up to k posteriors. That each interval has the same width is an assumption; that posterior means are equally spaced within each interval will be a result. Because $b > 0$ can be an arbitrarily large number, I de-emphasize the interpretation of the price range $[\ell b, (\ell + 1)b]$ for $\ell \in \mathbb{Z}$; it is merely a theoretical construct. In contrast, $\frac{b}{k} \equiv \varepsilon$ will map to the attention parameter from the example in Section 2.2 in equilibrium.

Ex ante, I assume that traders minimize expected errors arising from coarseness:

$$\min_{\mathcal{P}_1,...,\mathcal{P}_k} \int_{\ell b}^{(\ell+1)b} \min\{|\mathcal{P}_1 - \mathbb{E}^{mar}[p_{t+1}|p]|, ..., |\mathcal{P}_k - \mathbb{E}^{mar}[p_{t+1}|p]|\} g^{mar}(p) dp \tag{4}$$

for each $\ell \in \mathbb{Z}$. In (4), $\mathcal{P}_1, ..., \mathcal{P}_k$ denote feasible posterior means, $[\ell b, (\ell + 1)b]$ denotes the artificial price range, and the minimum inside of the integral represents the error arising from the coarse expectation. $\mathbb{E}^{mar}[p_{t+1}|p]$ denotes conditional price expectations and $g^{mar}(p)$ denotes the unconditional ergodic price density, both from the martingale

[9] See Gul et al. (2017) for a discussion about attention cost functions that are consistent with this type of restriction.

[10] An alternative interpretation is that prices are defined in natural logs (that is, $p_t = ln(\tilde{p}_t)$ for some underlying $\tilde{p}_t$).

economy from Section 2.1. To denote martingale, I use the superscript *mar*.

Implicitly, I have the following timing in mind when setting up ex ante problem (4). For periods *before* my economy begins, $t \in \{-1, -2, -3, ...\}$, my economy is populated only by exogenous noise traders and the market maker. Based on dynamics described in Section 2.1, I can calculate $\mathbb{E}^{mar}[p_{t+1}|p] = p$ and the ergodic price density, which solves

$$g^{mar}(x) = \int_{-\infty}^{\infty} dF^{mar}(x|p)g^{mar}(p)dp \tag{5}$$

where $F^{mar}(x|p_t)$ is the conditional distribution of p_{t+1} given p_t described by pricing equation (1). Condition (5) says that when prices in any given period are distributed according to $g^{mar}(p)$, prices in the next period will, again, be distributed according to $g^{mar}(p)$. The only nonstandard part of this ergodic price density is the range of integration. Because I have partitioned my price space into $[\ell b, (\ell + 1)b]$ for $\ell \in \mathbb{Z}$, I will focus on $g^{mar}(p)$ that integrate to one on each cell of this partition. In this sense, the $g^{mar}(p)$ employed in program (4) is an actual density.

The coarse Bayesian understands that her ex ante choice of posterior means will impact her trading payoffs throughout her infinite lifetime. (4) posits that she selects the best possible posterior means in the sense of incurring the smallest expected errors from her coarsening, subject to the information she has at $t = 0$: historically observed values of $\mathbb{E}^{mar}[p_{t+1}|p]$ and $g^{mar}(p)$. Importantly, she does not take into account the impact of her market entry on these objects, $\mathbb{E}^{mar}[p_{t+1}|p]$ and $g^{mar}(p)$; she simply uses the information available to her at $t = 0$. In contrast to the current setup, if the trader were able to re-optimize her attention choice every period, her solution would look very different: intuitively, she would want to concentrate posterior means around the current price, p_t.

To get a sense for problem (4), let $k = 4$, $\ell = 0$, $\mathbb{E}^{mar}[p_{t+1}|p] = p$, and let $g^{mar}(p)$ be the density of the Uniform$[0, b]$ distribution. It should be intuitive that the trader should optimally select an *equally* spaced set of posterior means, $\{\varepsilon/2, 3\varepsilon/2, 5\varepsilon/2, 7\varepsilon/2\}$; see Figure 2 for a graph of the error function (the function inside of the integrals in (4)) when posterior means are selected in this way.[11] This example motivates the choice of

[11] While it is not shown in Figure 2, the distance between the last point in the interval $[0, b]$ and the

equally-spaced posterior means in the example from Section 2.2.

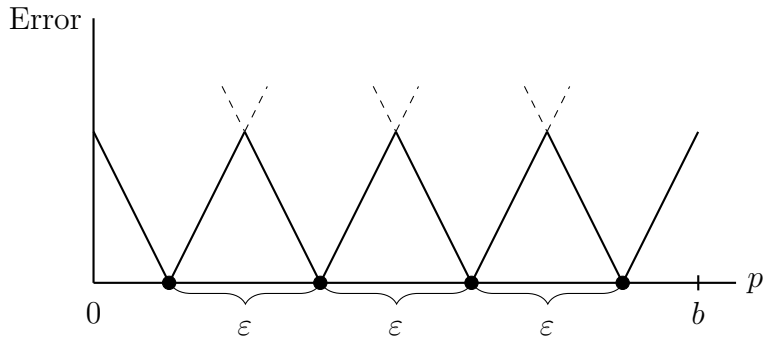

Figure 2: Error Arising from Coarse Expectations

To summarize, attention decisions are made early on in life ($t = 0$), while trading decisions are made later on throughout life ($t \geq 0$). These two problems—ex ante rounding and ex post trading—are not only solved separately from one another in my model, but they are typically understood to use different parts of the brain. In the words of Kahneman (2011), (ex ante) rounding is a System 1 problem; these are automatic, fast, effortless, and subconscious. (Ex post) trading is a System 2 problem; these are deductive, slow, effortful, and self-aware.

2.4 Equilibrium Definition

Given p_0, equilibrium consists of demand functions $x(p_t)$, a coarse expectation $\tilde{\mathbb{E}}[p_{t+1}|p_t]$, a coarse variance $\tilde{\text{Var}}(p_{t+1} - p_t|p_t)$, a conditional price distribution $F(x|p_t) \equiv Pr(p_{t+1} \leq x|p_t)$ and posterior means $\mathcal{P}$ such that, for all $t \geq 0$,

(a) $\mathcal{P} \cap [\ell b, (\ell + 1)b]$ solves (4) for all $\ell \in \mathbb{Z}$.

(b) Given $\mathcal{P}$ and $F(x|p_t)$, traders calculate $\tilde{\mathbb{E}}[p_{t+1}|p_t]$ using (3) and $\tilde{\text{Var}}(p_{t+1} - p_t|p_t) = \text{Var}(p_{t+1} - p_t|p_t)$.

(c) Given $\tilde{\mathbb{E}}[p_{t+1}|p_t]$ and $\tilde{\text{Var}}(p_{t+1} - p_t|p_t)$, demand functions $x(p_t)$ satisfy (2).

first point in the interval $[b, 2b]$ is also ε.

(d) Given $x(p_t)$, $F(x|p_t)$ is formed according to (6):

$$p_{t+1} = p_t + c[u_t + x(p_t)] \tag{6}$$

Condition (a) says that feasible posterior means, when restricted to the range $[\ell b, (\ell+1)b]$ in which traders are allotted k posteriors, are optimally chosen given the historical price distributions available to the trader at time $t = 0$. Condition (b) requires that traders formulate coarse expectations using the true equilibrium price distribution and posterior means that solve the optimal attention allocation problem. Condition (b) also leverages the fact that the true price variance is constant in p_t, because noise only enters the model through the noise trader. Then, irrespective of how the posteriors are partitioned, a coarse Bayesian should hold the correct belief over the price variance. Condition (c) says that demand maximizes expected utility, and Condition (d) says that prices are made according to the market maker. Importantly, the conditional price distribution, $F(x|p_t)$, not only accounts for noise trading, u_t, but also coarse Bayesian trading, $x(p_t)$. Note that p_{t+1} is a random variable due to noise trading, u_t.[12]

I close this section with a brief discussion about modeling assumptions. The empirical reason for avoiding a market clearing setup—and, instead, introducing the market maker in Section 2.1—is that technical traders map better to retail traders than institutional traders, and papers like Barber et al. (2009) discuss the extensive use of market orders by retail traders. However, there is also a technical reason why I am unable to employ a standard market-clearing setup: there would be an infinite multiplicity of market-clearing prices according to demands depicted in Figure 1.

An alternative modeling assumption to attention problem (4) would be to use the equilibrium $F(x|p)$ to calculate both $\mathbb{E}[p_{t+1}|p]$ and the ergodic density of the model *incorporating coarse Bayesian trading*, which solves

$$g(x) = \int_{-\infty}^{\infty} dF(x|p)g(p)dp, \tag{7}$$

with the understanding that $g(p)$ integrates to one over $[\ell b, (\ell+1)b]$ for $\ell \in \mathbb{Z}$, and using

[12] More formally, I assume that the realization of u_t is not known to coarse Bayesians at the time of trade in period t.

these objects in lieu of $\mathbb{E}^{mar}[p_{t+1}|p]$ and $g^{mar}(p)$ in the attention problem (4). I choose not to take this approach for two reasons: i) tractability of the model, but also ii) such a calculation would require a colossal amount of information on the part of the (already attention-constrained) trader.

3 Equilibrium Characterization

As alluded to in Section 2.4, my equilibrium concept is tractable because Condition (a), the ex ante problem, can be solved independently of Conditions (b) through (d), the ex post problem. I begin with the ex ante problem.

3.1 Attention Problem

The attention problem (4) first requires an understanding of the benchmark economy described in Section 2.1. Because the only thing determining prices is symmetric noise trading, prices will be a random walk. And when conditional prices are a random walk, the unconditional ergodic price density should intuitively be Uniform. My first lemma confirms these claims. All proofs are in Appendix A.

Lemma 1. ***(Uniform Density)*** *Consider the economy with no coarse Bayesians, described by pricing equation* (1)*. Then*

(a) One ergodic density that solves (5) *is constant over all x, and*

(b) The density of p_{t+1} conditional on $p_{t+1} \in [\ell b, (\ell+1)b]$ converges pointwise to this constant function as $t \to \infty$ for any initial p_0 and $\ell \in \mathbb{Z}$.

Part (a) confirms the Uniform ergodic density, and part (b) ensures that the economy always approaches this Uniform limit dynamically. Important to Lemma 1 is that a noise trading density $h(u)$ exists; without absolute continuity, one could imagine an ergodic distribution with a sequence of discrete points as its support. With the Uniform (historical) ergodic density confirmed, my first theorem characterizes the solution to the attention problem (4).

Theorem 1. ***(Attention Problem)*** *The unique solution to the following attention problem,*

$$\min_{\mathcal{P}_1,...,\mathcal{P}_k} \int_{\ell b}^{(\ell+1)b} \min\{|\mathcal{P}_1 - p|, ..., |\mathcal{P}_k - p|\} \frac{1}{b} dp, \tag{8}$$

is $\mathcal{P}_i = \ell b + \varepsilon(i-1) + \varepsilon/2$.

Theorem 1 states that the solution to the attention program (8) is unique and equally-spaced. Note that the program (8) is the original program (4), but with $\mathbb{E}^{mar}[p_{t+1}|p] = p$, which follows directly from pricing equation (1), and $g^{mar}(p) = \frac{1}{b}$, which is the density of the Uniform distribution and guaranteed as a fixed point of (5) by Lemma 1. The intuition behind Theorem 1 is exactly that of Figure 2, justifying the use of the Uniform distribution and $\mathbb{E}^{mar}[p_{t+1}|p] = p$ in that example. Because Theorem 1 holds for any $\ell \in \mathbb{Z}$, I can "stitch together" the entire sequence of (optimal) posterior means, $\mathcal{P} = \{\varepsilon/2 + \varepsilon i\}_{i \in \mathbb{Z}}$.

3.2 Conditional Prices

With the attention problem (4) solved, I move on to equilibrium Conditions (b) through (d). I begin with the simplest calculation, which is of the equilibrium price variance. Recall that this value is constant in p_t due to the fact that price noise is derived from (inelastic) noise trading. To see this formally, consider the pricing equation (6). Taking the variance of both LHS and RHS, I have that

$$\text{Var}(p_{t+1} - p_t | p_t) = c^2 \text{Var}(u_t) \tag{9}$$

so that noise trader volatility is directly passed on to prices. The coarse Bayesian, who holds correct beliefs about the price variance, uses this value in their utility program.

Next I consider the first moment of prices, which can be calculated in a similar way to the equilibrium price variance above. Taking the mean of the LHS and RHS of the pricing rule (6),

$$\mathbb{E}[p_{t+1} - p_t | p_t] = cx(p_t) = c\frac{\tilde{\mathbb{E}}[p_{t+1}|p_t] - p_t}{\rho c^2 \text{Var}(u_t) + \tau} \tag{10}$$

where the first equality follows from noise trading having zero mean, and the second

equality follows from the variance calculation (9). Unlike (9), equation (10) involves an equilibrium calculation because the RHS depends on the coarse expectation (3), which in turn depends on $\mathbb{E}[p_{t+1}|p_t]$. My second lemma characterizes one solution to (10).

Lemma 2. ***(Coarse Expectation)*** *There exists an equilibrium where the coarse expectation* (3) *equals*

$$\hat{\mathbb{E}}[p_{t+1}|p_t] \in argmin_p |p - p_t| \;\; s.t. \;\; p \in \mathcal{P}$$

The difference between this formula and (3) is that p_t is used in lieu of $\mathbb{E}[p_{t+1}|p_t]$. This lemma is not only useful algebraically, but it also preserves the intuition developed in Figure 1 (which relied on $\mathbb{E}[p_{t+1}|p_t] = p_t$]). To see how Lemma 2 works, consider the coarse expectation calculated as in the lemma. If the inequality $|\hat{\mathbb{E}}[p_{t+1}|p_t] - \mathbb{E}[p_{t+1}|p_t]| \leq \varepsilon/2$ holds true for all p_t, then the trader would have selected the *same* coarse expectation even when considering the true equilibrium price expectation in its calculation; or in other words, $\hat{\mathbb{E}}[p_{t+1}|p_t] = \tilde{\mathbb{E}}[p_{t+1}|p_t]$. Consider the following decomposition of the desired inequality:

$$|\hat{\mathbb{E}}[p_{t+1}|p_t] - \mathbb{E}[p_{t+1}|p_t]| = |\underbrace{(\hat{\mathbb{E}}[p_{t+1}|p_t] - p_t)}_{Coarseness} + \underbrace{(p_t - \mathbb{E}[p_{t+1}|p_t])}_{Non-martingale}| \leq \varepsilon/2 \tag{11}$$

Because posterior means $\mathcal{P}$ have spacing ε, the coarse expectation calculated as in Lemma 2, $\hat{\mathbb{E}}[p_{t+1}|p_t]$, satisfies $|\hat{\mathbb{E}}[p_{t+1}|p_t] - p_t| \leq \varepsilon/2$ by construction. Then, for (11) to hold, it must be that errors attributed to prices not being a martingale *cancel out* with errors attributed to coarseness.

The intuition on why these two sources of error cancel is as follows. If p_t is located immediately below (above) $\hat{\mathbb{E}}[p_{t+1}|p_t]$, the coarseness term in the decomposition (11) is positive (negative). For the non-martingale term in the decomposition (11), consider equation (10). Because I am considering an equilibrium where $\tilde{\mathbb{E}}[p_{t+1}|p_t] = \hat{\mathbb{E}}[p_{t+1}|p_t]$, I substitute this equality into the RHS of (10). Then, all of the intuition from Figure 1 goes through: if p_t is located immediately below (above) $\hat{\mathbb{E}}[p_{t+1}|p_t]$, the coarse Bayesian buys (sells). According to (10), then, prices are a submartingale (supermartingale) so actual price expectations are higher (lower) than p_t. This diminishes the size of the total error, confirming the earlier use of the equality $\tilde{\mathbb{E}}[p_{t+1}|p_t] = \hat{\mathbb{E}}[p_{t+1}|p_t]$.

What if errors from coarseness and prices not being a martingale from the decomposition (11) still work in opposite directions, but the non-martingale term is simply too large in magnitude? A subtle parameter restriction rules out this kind of price behavior: because zero expected profits by the market maker imposes that $c = \tau/2$, the term that multiplies the coarse expectation in (10), $\frac{c}{\rho c^2 \mathrm{Var}(u_t)+\tau}$, is always less than one. This keeps the non-martingale term small in magnitude. Altogether, Lemma 2 says: because there is not too much coarse Bayesian trading, prices are close enough to a martingale that a coarse Bayesian cannot tell the difference.

Note that Lemma 2 potentially allows for multiplicity of the following kind: a belief of rising prices leads to trades that confirm such a rising price. This kind of multiplicity is a necessary consequence of a setting without dividends, which I refrain from adding to the model to maintain the paper's focus on technical trading.[13] Interestingly, the multiplicity is coarsening-specific. To see why this is, consider what would happen if, instead of appending coarse Bayesians, I appended only rational mean-variance traders to the martingale economy from Section 2.1. Then, an equation analogous to (10) obtains, but with a non-coarse expectation on the RHS. But I have previously noted that the term that multiplies this expectation, $\frac{c}{\rho c^2 \mathrm{Var}(u_t)+\tau}$, is strictly less than one. In words, a trader who believes that actual prices will rise (fall) trades in such a way that prices rise (fall) but by a *smaller* magnitude. The only equilibrium solution to this hypothetical situation is to uniquely set $\mathbb{E}[p_{t+1} - p_t|p_t] = 0$. In contrast, small changes to $\mathbb{E}[p_{t+1}|p_t]$ can lead to large changes in $\tilde{\mathbb{E}}[p_{t+1}|p_t]$, so this logic breaks down with coarse traders.

The reason for selecting this particular equilibrium in Lemma 2 is similar to the reason for using $g^{mar}(p)$ in the attention problem (4). There, I assumed that traders used historical price distributions when solving their attention problem because it was unrealistic for attention-constrained traders to hypothesize previously-unobserved fixed points, $g(p)$, that solve (7). Unlike the attention problem, the situation here is slightly different because traders re-trade each period (and, hence, it would be incorrect of traders to continue to use $\mathbb{E}^{mar}[p_{t+1}|p_t] = p_t$ for $t \geq 0$). However, the equilibrium I have selected allows traders to continue to believe (or, more precisely, act as if they believe) that $\mathbb{E}[p_{t+1}|p_t] = p_t$ due to their coarsening, despite it not being the true process for

[13] Papers like Han and Kumar (2013) find that retail traders—who map well to coarse Bayesians—are much more likely (than institutional ones) to buy stocks with no dividends.

prices. These modeling choices are made to avoid Simon's infinite regress problem: that optimizing the allocation of thinking cost can be even more complex than the original problem.

3.3 Simulations

To motivate my theoretical characterization of support and resistance levels in upcoming Section 4, I run a simulation of the equilibrium described in Sections 3.1 and 3.2. In this simulation, I let the price elasticity $c = 1$, the spacing $\varepsilon = 100$, risk aversion $\rho = 1$, transaction costs $\tau = 2$, and the noise trading distribution $u_t \sim Normal(0, 25)$. Note that these parameters have been chosen in such a way that the market maker makes zero expected profits. Figure 3 displays a simulation of the first 1000 time periods in my economy, starting at price $p_0 = 1000$.

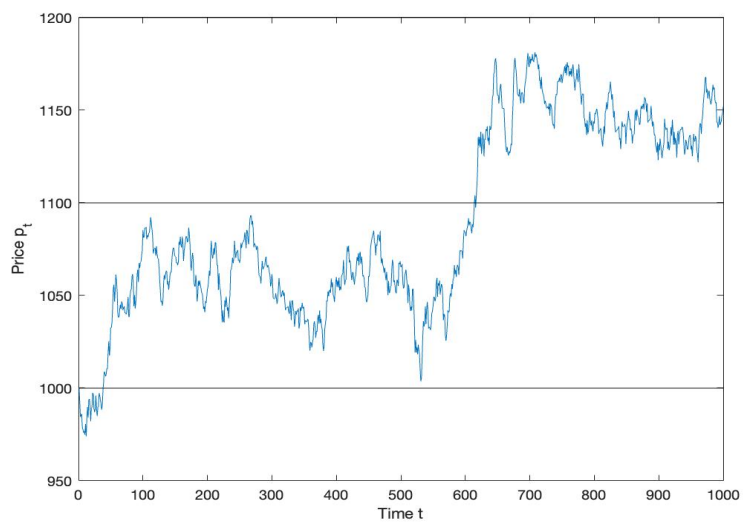

Figure 3: Economy Simulation

To understand this graphic, consider the coarse Bayesian demand function. It has period $\varepsilon = 100$, with discontinuities (where the coarse Bayesian shifts from one posterior mean to the next) denoted by the horizontal black lines in Figure 3. Let me call a horizontal white band, demarcated by two black lines in Figure 3, as a regime. As prices

approach the top (bottom) of any given regime, the coarse Bayesian believes that the asset is over-priced (under-priced) and sells (buys). While it is not always the case due to the presence of price noise, this tends to push the price back into the original regime.

Next, I continue to let the simulation from Figure 3 run for 1,000,000 time periods and plot the ergodic density of $\text{mod}(p, \varepsilon)$ in Figure 4, where $\text{mod}(\cdot)$ denotes the modulo function. While it may be unclear why this particular ergodic density is of interest, formal definitions of support and resistance levels from upcoming Section 4 will make use of this distribution. At an intuitive level, when prices deflect off of support and resistance levels in Figure 3, one might expect this ergodic density to put more (less) probability mass in the center (edges) of the interval $p_t \in [0, 100]$ in Figure 4.

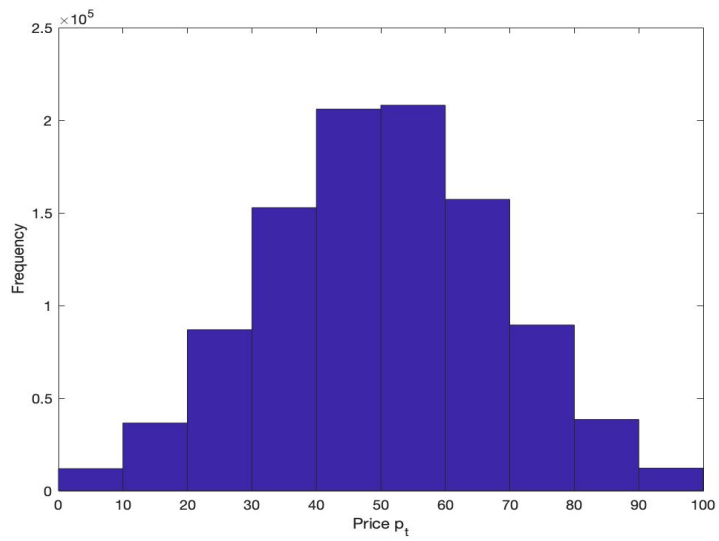

Figure 4: Histogram of Ergodic Density of $\text{mod}(p, \varepsilon)$

Two features of the ergodic density in Figure 4 that I emphasize are its symmetry and single-peakedness. While symmetry will be guaranteed in my next lemma with no additional assumptions, that the density puts more mass in the center of the range $p_t \in [0, 100]$ is a common feature, but not a guaranteed property, in my model.

4 Support and Resistance

While Figures 3 and 4 provided informal evidence of support and resistance levels, here I provide and verify (for my economy) formal definitions of support and resistance levels. The starting point will be the ergodic density of $\text{mod}(p, \varepsilon)$, which I will denote $g^{mod}(p)$.

4.1 Existence

My final lemma shows that the ergodic density of $\text{mod}(p, \varepsilon)$, $g^{mod}(p)$, satisfying the following fixed point condition (12) not only exists, but is symmetric:

$$g^{mod}(x) = \int_0^{\varepsilon} f^{mod}(x|\mathcal{P}) g^{mod}(\mathcal{P}) d\mathcal{P} \tag{12}$$

where $f^{mod}(x|\mathcal{P})$ denotes the conditional distribution of $\text{mod}(p_{t+1}, \varepsilon)$ given $\mathcal{P} \equiv \text{mod}(p, \varepsilon)$. First, note that the range of integration in (12) is $[0, \varepsilon]$, which is the support of $\text{mod}(p_{t+1}, \varepsilon)$. Second, it is not obvious that conditioning on $\mathcal{P} \equiv \text{mod}(p, \varepsilon)$, as opposed to p, is sufficient to determine the distribution $f^{mod}(x|\mathcal{P})$. While this is formally verified in Lemma 3, this is intuitively because coarse demand $x(p)$ is measurable with respect to $\mathcal{P}$. That is, $x(p) = x(\mathcal{P})$.

Lemma 3. ***(Ergodic Density)*** *Assume that the equilibrium described in Lemmas 1 and 2 is selected. Then a $g^{mod}(p)$ satisfying* (12)

(a) exists, and

(b) $g^{mod}(\varepsilon/2 + \delta) = g^{mod}(\varepsilon/2 - \delta)$ for $0 \leq \delta \leq \varepsilon/2$.

Lemma 3 is an existence proof. Before exploring fixed points of (12), I first guarantee an important condition pertaining to the conditional density of mod prices:

$$f^{mod}(x|\varepsilon/2 - \delta) = f^{mod}(\varepsilon - x|\varepsilon/2 + \delta) \text{ for } 0 \leq \delta < \varepsilon/2,$$

which I call *conditional symmetry*, and the equality above holds for all x. Conditional symmetry says that a reflection of the current (mod) price over the point $\varepsilon/2$ reflects the

conditional (mod) price density also over the point $\varepsilon/2$. Intuitively speaking, conditional symmetry follows from i) symmetry in the coarse Bayesian's problem and ii) assumed symmetry of the noise trading process. I then verify that $g^{mod}(p)$ "inherits" this property in the sense of Lemma 3 part (b). While Lemma 3 does not rule out multiple fixed points of (12), numerical simulations suggest that the symmetric one is approached dynamically.

4.2 Definition

The main definition of support and resistance levels I use is from Donaldson and Kim (1993). In their empirical study with $\varepsilon = 100$, they verify the following property of the Dow Jones Industrial Average:

Definition 1. ***(Support and Resistance)*** *Prices are characterized by support and resistance levels if $Cov^{mod}(\mathbb{E}[p_{t+1}|p] - p, p) < 0$, where the conditional expectation (covariance) is taken with respect to $F(x|p)$ ($g^{mod}(p)$).*

Definition 1 explains the need for investigating the ergodic distribution of $\text{mod}(p, \varepsilon)$ in Figure 4 and Lemma 3: $g^{mod}(p)$ is used to define support and resistance levels. Taking one step back, Definition 1 highlights the connection between support/resistance and equally-spaced posterior means: equal spacing is what leads to a periodic demand function $x(p)$, which plays a crucial role in the existence argument for $g^{mod}(p)$ in Lemma 3. Next, consider the covariance in Definition 1. In the bottom (top) part of the interval $[0, \varepsilon]$, I argued that prices are a sub- (super-) martingale in Section 3.2. The sub- and super-martingale results, together, imply that the covariance in Definition 1 is negative.

Theorem 2. ***(Support and Resistance)*** *Assume that the equilibrium described in Lemmas 1 and 2 is selected. Furthermore, assume that the ergodic density $g^{mod}(p)$ described in Lemma 3 is selected. Then prices are characterized by support and resistance levels in the sense of Definition 1.*

Consider how Definition 1 relates back to the intuitive notion of support and resistance: price levels where prices tend to deflect, but move more so than otherwise warranted if not. Definition 1 ensures that prices tend to rise in $p \in [0, \varepsilon/2]$ and fall in $p \in [\varepsilon/2, \varepsilon]$, so it is *as-if* prices deflect off of support level 0 from above and resistance

level ε from below. Furthermore, consider when prices break the resistance level ε from below. Due to the modulo function used in Definition 1, prices tend to *continue to rise* when $p \in [\varepsilon, 3\varepsilon/2]$. In this sense, prices move more so than otherwise warranted immediately after the range is broken. Indeed, standard range-break trading strategies prescribe buying (selling) when a resistance (support) level is broken, which is consistent with the returns process generated in my model.

To appreciate the equilibrium nature of calculations involved in Theorem 2, consider the following equality:

$$\int_0^{\varepsilon} (\mathbb{E}[p_{t+1}|p] - p) g^{mod}(p) dp = 0,$$

which holds true in my setting. Although conditional prices are not a martingale, this equation says that—unconditionally—prices neither rise nor fall. The properties required for this equality to hold are conditional symmetry of $f(x|p)$ and symmetry of $g^{mod}(p)$. While the latter is guaranteed by Lemma 3, the former refers to the equilibrium conditional price distribution $f(x|p)$, not $f^{mod}(x|p)$ investigated in Lemma 3. Note that the equality above is not just for the sake of explanation; it is directly used in the covariance calculation of Theorem 2.

Donaldson and Kim (1993) not only empirically verify Definition 1, but also a second property of the Dow Jones Industrial Average: in their words, that the ergodic density $g^{mod}(p)$ is "hump-shaped." They verify this in two ways:

Definition 2. ***(Support and Resistance)*** *Prices are characterized by support and resistance levels if*

(a) $g^{mod}(0) \leq g^{mod}(p)$ *and* $g^{mod}(\varepsilon) \leq g^{mod}(p)$ *for all* $p \in [0, \varepsilon]$*, and*

(b) $g^{mod}(p)$ *is concave for* $p \in [0, \varepsilon]$*,*

where $g^{mod}(p)$ *denotes the ergodic density of* $mod(p, \varepsilon)$*.*

Condition (a) says that the ergodic density has minima at the edges of the range $p \in [0, \varepsilon]$; Condition (b) is self-explanatory. Definition 2 comes with two complications. First, the reason for checking these properties in Donaldson and Kim (1993) is based on foreign-exchange based (not equity based) theories of support and resistance levels.

Even within that literature, there are conflicting views as to whether the ergodic density is hump-shaped: Krugman (1991) argues that the distribution is U-shaped while Bertola and Caballero (1992) argue that it is hump-shaped. Second, Figure 4 might suggest that my model is always characterized by a hump-shaped $g^{mod}(p)$. But in fact, the hump-shaped property of $g^{mod}(p)$ imposes significant restrictions on the noise trading distribution $h(u)$, which I have kept fairly general so far.

My next example showcases this last claim. I make two changes to the parameter values from Section 3.3, but otherwise keep the example intact. For no other reason than to make the failure of Definition 2 easier to discern visually, I reduce $\varepsilon = 10$. For the noise trading distribution, instead of the $Normal(0, 25)$ distribution, I assume that with a 50% chance, $u_t \sim$Uniform$[-5, -4]$, and with the other 50% chance, $u_t \sim$Uniform$[4, 5]$. This distribution i) has a distance of $\varepsilon = 10$ from one end of the support to the other, and ii) mimics a random walk taking on only values in $\{-5, 5\}$ while maintaining absolute continuity.[14] The ergodic density of mod(p, ε) after 1,000,000 simulations is shown in Figure 5.

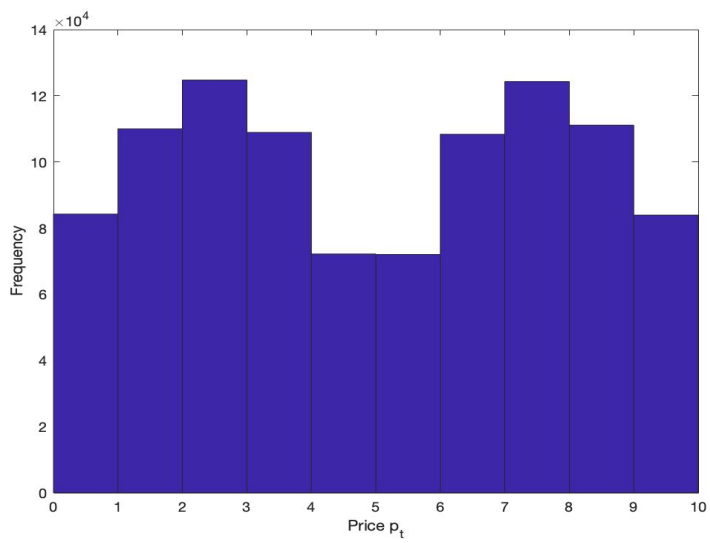

Figure 5: Histogram of Non-Hump-Shaped $g^{mod}(p)$

There are two competing forces at work in Figure 5. First, coarse Bayesian trading

[14] Absolute continuity is assumed in the model because a density, $h(u)$, is assumed to exist.

pushes the price p_{t+1} up (down) when the price is in $p_t \in [0, 5]$ ($p_t \in [5, 10]$). This tends to push prices toward the horizontal *center* of the graph in Figure 5. Second, when prices p_t are near this center point, noise trading tends to push next-period-prices p_{t+1} to the edge of the price range $p_t \in [0, 10]$. Because coarse Bayesians repel prices from the edges and noise trading repels prices from the center of the graph, neither center nor edges receive the modal probability mass in the ergodic distribution. In contrast, these two competing forces *aligned* in the simulation from Section 3.3.[15]

One way to summarize this section is that the empirical observation by Donaldson and Kim (1993) *disciplines* the family of admissible noise trading distributions. However, I do not pursue this interpretation here because the rest of my theory puts no restrictions on the noise trading density, $h(u)$, beyond its symmetry (and several technical properties). A second way to summarize this section is to endorse Definition 1 over Definition 2. At the very least, I emphasize the fact that Definition 2 imposes additional structure on other non-coarse aspects of the economy—that is, $h(u)$ in my model—which Definition 1 does not.

4.3 Robustness

An important question about this theory is whether it is robust to arbitrage from rational traders. More precisely, consider adding a mass of rational traders to the economy: they would buy (sell) at prices p_t for which the first moment is positive (negative). The point of this discussion is that rational arbitrage exacerbates, not alleviates, the support and resistance problem. This relates to a literature on the limits to arbitrage.[16] My only corollary makes this point more formal.

Corollary 1. ***(Rational Arbitrage)*** *Fix the price elasticity $c = \tau/2$ across two economies: the one described in Section 2 and the second with an additional mass of identical, myopic, rational, mean-variance traders. Assume that the equilibrium described in Lemmas*

[15] While the example in Figure 5 was selected for its clarity, other examples can be constructed where i) the support of $h(u)$ is convex, ii) $h(u)$ is concave along that support, and iii) $h(-\mathcal{C}) \leq h(u)$ and $h(\mathcal{C}) \leq h(u)$ for all u in the support $[-\mathcal{C}, \mathcal{C}]$, yet Definition 2 parts (a) and (b) both fail.

[16] For example, see De Long et al. (1990).

1 and 2 is selected in both economies. Then

$$\begin{aligned} &\overline{\mathbb{E}}[p_{t+1}|p] - p > \mathbb{E}[p_{t+1}|p] - p > 0 \textit{ for } p \in (0, \varepsilon/2), \textit{ and} \\ &\overline{\mathbb{E}}[p_{t+1}|p] - p < \mathbb{E}[p_{t+1}|p] - p < 0 \textit{ for } p \in (\varepsilon/2, \varepsilon) \end{aligned} \tag{13}$$

where price moments with (without) a bar denote those of the economy with rational traders (the original economy).

In Corollary 1, rational traders buy (sell) when prices are a submartingale (supermartingale), exacerbating the problem of support and resistance as described in (13). Furthermore, because rational arbitrageurs trade based on *periodic* price moments, their quantities demanded are also periodic in p_t and condition (13) can be extended to any price range. Consider the need for fixing c: with additional traders in the economy, the zero-profit condition may fail with $c = \tau/2$. As Appendix B shows, this failure has nothing to do with rational versus coarse traders but, instead, with the change in total mass of traders in the economy. Because a smaller price elasticity c can work against the effect described in (13), I am forced to shut down this channel in Corollary 1.

Taking expectations of the pricing rule (6), including arbitrageurs, and plugging in variance condition (9),

$$\overline{\mathbb{E}}[p_{t+1}|p_t] - p_t = c\frac{\tilde{\mathbb{E}}[p_{t+1}|p_t] - p_t]}{\rho c^2 \mathrm{Var}(u_t) + \tau} + c\frac{\overline{\mathbb{E}}[p_{t+1}|p_t] - p_t}{\hat{\rho} c^2 \mathrm{Var}(u_t) + \tau}$$

where $\hat{\rho} > 0$ denotes the risk aversion coefficient of rational traders. Recursively plugging in $\overline{\mathbb{E}}[p_{t+1}|p_t]$ into the RHS of this equation, I require that this geometric series converges. But for similar reasons to those laid out in Section 3.2, $\frac{c}{\hat{\rho} c^2 \mathrm{Var}(u_t) + \tau} < 1$ and the series indeed converges. This geometric series is a useful way to think about the amplification provided by rational arbitrageurs: they buy (sell), which makes prices rise (fall), which makes them buy (sell) more, and so on. To summarize, my economy can be characterized by very little coarse trading. Instead, it is the *knowledge* of support and resistance levels — by rational arbitrageurs — that can generate significant support and resistance levels in equilibrium prices.

An alternative to testing condition (13) would be to check if $\mathrm{Cov}^{mod}(\mathbb{E}[p_{t+1}|p] - p, p)$

from Definition 1 decreases with the addition of rational traders. It turns out that this last statement need not be true for the following reason. Rational traders trade in the same way as coarse Bayesians; they buy (sell) when coarse Bayesians buy (sell). Then adding rational traders is *as-if* I added more coarse Bayesians to the economy, which tends to tighten the ergodic density $g^{mod}(p)$ (i.e. lower its variance within the range $[0, \varepsilon]$). This can offset the effect captured in (13) when calculating $\text{Cov}^{mod}(\mathbb{E}[p_{t+1}|p] - p, p)$.

While this section focused on an extreme form of heterogeneity, a rational group and an inattentive group, similar conclusions hold with a more attentive and less attentive group. First consider what it means to be less attentive: I fix the theoretical price range $[\ell b, (\ell + 1)b]$ for $\ell \in \mathbb{Z}$, but allow one attentive group to have more posteriors, k, per range and another inattentive group to have fewer posteriors, $k' < k$. Theorem 1 then implies that $\varepsilon < \varepsilon'$. What makes this exercise interesting is the failure of Lemma 2: due to the large magnitude of trading by the inattentive group, the attentive group will—for reasonable parameter values—realize that prices are not a martingale. They buy (sell) more than what is simply prescribed by their coarseness when prices are expected to rise (fall), providing similar amplification to that described in Corollary 1.

5 Conclusion

There were two main takeaways from the paper. First, when traders faced an ex ante attention problem of choosing their optimal set of posterior means, they chose an equally-spaced price grid to minimize expected errors. This led to demand functions with both a conventional downward-sloping component, as well as a speculative upward-discontinuity where higher prices in period t were used as an indication of higher prices in period $(t+1)$. Second, equilibrium prices were characterized by support and resistance levels, or price levels where prices tend to deflect (but move more so than otherwise warranted if not). When prices rose (fell) to the top (bottom) of a regime, coarse Bayesians believed that the asset was overpriced (underpriced); this created downward (upward) price pressure, pushing prices back into said regime.

While I have introduced my model in the context of equity markets, the mechanism is general enough to apply to other financial markets that include retail traders and

market makers. In particular, there are well-established empirical literatures focused on support and resistance levels in foreign exchange markets (see Qi and Wu (2006)) and in commodities markets (see Aggarwal and Lucey (2007)), and also growing evidence that support and resistance levels are prevalent in cryptocurrency markets. For example, Hudson and Urquhart (2021) find significant predictability and profitability in several cryptocurrencies using support-resistance trading rules.

A Proofs

A.1 Proof of Lemma 1

With no coarse Bayesians, the conditional distribution of p_{t+1} is given by $Pr(p_{t+1} \leq x|p) = Pr(p + cu_t \leq x) = Pr(u_t \leq [x-p]/c)$ with density $h([x-p]/c)/c$. Consider the fixed point condition (5) with a Uniform density $g(p) = 1/b$ on the RHS. Furthermore, consider this fixed point evaluated at two points, x versus y. At y,

$$g(y) = \int_{-\infty}^{\infty} h([y-p]/c)\frac{1}{cb}dp = \int_{-\infty}^{\infty} h([x-p']/c)\frac{1}{cb}dp' = g(x)$$

where the second equality follows from a change of variable, $y-p = x-p'$. This line of reasoning shows that $g(x)$ is constant.

For part (b) of the lemma, I calculate $p_{t+1} = \sum_{s=0}^{t} cu_s + p_0$ by recursively substituting equation (1). I let $f_t(x)$ denote the density of $\mathcal{X}_t \equiv (cu_0 + ... + cu_t)/\sqrt{\text{Var}(cu_t)(t+1)}$. Importantly, this density exists for all t because $h(u)$ has a density (and hence, the density of the sum is given by the convolution formula). By the limit theorem for densities, $h(u)$ being bounded (with a finite second moment) is a sufficient condition to ensure that $f_t(x)$ converges uniformly to the standard Normal density as $t \to \infty$; call that limiting density $\varphi(x)$.[17] My interest is in the variable $\sqrt{\text{Var}(cu_t)(t+1)}\mathcal{X}_t + p_0$, which has density $f_t([x - p_0]/\sqrt{\text{Var}(cu_t)(t+1)})/\sqrt{\text{Var}(cu_t)(t+1)}$. Take any two points $y, z \in [\ell b, (\ell+1)b]$ and consider the ratio of densities at those points:

$$\frac{f_t([y-p_0]/\sqrt{\text{Var}(cu_t)(t+1)})}{f_t([z-p_0]/\sqrt{\text{Var}(cu_t)(t+1)})} \to \frac{\varphi(0)}{\varphi(0)} = 1 \tag{14}$$

where the limit taken is $t \to \infty$. To show the convergence step more carefully, I claim that, for every $\varepsilon > 0$ there is T such that $t > T$ implies

$$|f_t([y-p_0]/\sqrt{\text{Var}(cu_t)(t+1)}) - \varphi(0)| < \varepsilon$$

By the triangle inequality,

$$\begin{aligned}
&|f_t([y-p_0]/\sqrt{\text{Var}(cu_t)(t+1)}) - \varphi(0)| \\
\leq &|f_t([y-p_0]/\sqrt{\text{Var}(cu_t)(t+1)}) - \varphi([y-p_0]/\sqrt{\text{Var}(cu_t)(t+1)})| \\
+&|\varphi([y-p_0]/\sqrt{\text{Var}(cu_t)(t+1)}) - \varphi(0)|
\end{aligned}$$

For the first term on the RHS, by uniform convergence of the density f_t, there is T_1 such that this first term is less than $\varepsilon/2$ for $t > T_1$. For the second term on the RHS, by continuity of the standard Normal density at zero, there is T_2 such that this second term is less than $\varepsilon/2$ for

[17] See Gnedenko and Kolmogorov (1954) chapter 8.

$t > T_2$. Then take $T = \max\{T_1, T_2\}$. Note that T_2 depends on y, and hence T depends on y.

Back to my original line of reasoning, I have shown that the limiting density is constant (because the ratio evaluated at y versus z is one). Note that $\sqrt{\text{Var}(cu_t)(t+1)}\mathcal{X}_t + p_0$, conditional on being in the range $[\ell b, (\ell+1)b]$, must also have a constant density because any normalization required by Bayes' Rule cancels when the ratio (14) is considered. □

A.2 Proof of Theorem 1

Ignoring positive constants, the attention problem is

$$\min_{\mathcal{P}_1,\ldots,\mathcal{P}_k} \int_{\ell b}^{(\ell+1)b} \min\{|\mathcal{P}_1 - p|, \ldots, |\mathcal{P}_k - p|\} dp = \min_{\ell b \leq \mathcal{P}_1 \leq \ldots \leq \mathcal{P}_k \leq (\ell+1)b} \sum_{i=1}^{k} \int_{(\mathcal{P}_{i-1}+\mathcal{P}_i)/2}^{(\mathcal{P}_i+\mathcal{P}_{i+1})/2} |\mathcal{P}_i - p| dp$$

with the understanding that $(\mathcal{P}_0 + \mathcal{P}_1)/2 = \ell b$ and $(\mathcal{P}_k + \mathcal{P}_{k+1})/2 = (\ell+1)b$. Note that the ranges of integration follow from the fact that the active term in the inner minimum switches when

$$p - \mathcal{P}_i = \mathcal{P}_{i+1} - p \iff p = \frac{\mathcal{P}_i + \mathcal{P}_{i+1}}{2}$$

I take a first-order condition with respect to $\mathcal{P}_i$, guessing that inequality constraints do not bind:

$$\frac{1_{\{i\neq 1\}}}{2}\left|\frac{\mathcal{P}_{i-1} - \mathcal{P}_i}{2}\right| - \frac{1_{\{i\neq 1\}}}{2}\left|\frac{\mathcal{P}_i - \mathcal{P}_{i-1}}{2}\right| + \int_{(\mathcal{P}_{i-1}+\mathcal{P}_i)/2}^{\mathcal{P}_i} dp$$
$$- \int_{\mathcal{P}_i}^{(\mathcal{P}_i+\mathcal{P}_{i+1})/2} dp + \frac{1_{\{i\neq k\}}}{2}\left|\frac{\mathcal{P}_i - \mathcal{P}_{i+1}}{2}\right| - \frac{1_{\{i\neq k\}}}{2}\left|\frac{\mathcal{P}_{i+1} - \mathcal{P}_i}{2}\right| = 0$$

or, more simply,

$$\int_{(\mathcal{P}_{i-1}+\mathcal{P}_i)/2}^{\mathcal{P}_i} dp - \int_{\mathcal{P}_i}^{(\mathcal{P}_i+\mathcal{P}_{i+1})/2} dp = 0 \iff \mathcal{P}_i = \frac{\mathcal{P}_{i-1} + \mathcal{P}_{i+1}}{2}$$

when $1 < i < k$. Next consider the endpoints. For the lower endpoint $i = 1$, the first-order condition simplifies to

$$\mathcal{P}_1 - \ell b - \left(\frac{\mathcal{P}_1 + \mathcal{P}_2}{2} - \mathcal{P}_1\right) = 0 \iff \frac{3}{2}\mathcal{P}_1 = \ell b + \frac{\mathcal{P}_2}{2}$$

By similar logic, the upper endpoint $i = k$ is characterized by first-order condition

$$\mathcal{P}_k - \frac{\mathcal{P}_{k-1} + \mathcal{P}_k}{2} - ([\ell+1]b - \mathcal{P}_k) = 0 \iff \frac{3}{2}\mathcal{P}_k = (\ell+1)b + \frac{\mathcal{P}_{k-1}}{2}$$

This system is solved by $\mathcal{P}_i = \ell b + (i-1)\varepsilon + \varepsilon/2$ for $i = 1, .., k$. Note that my previous guess—that no inequality constraints bind—is satisfied and, hence, all multipliers are zero at this critical point by complementary slackness. The Hessian of the objective is (still ignoring positive constants)

$$\mathcal{H} = \begin{bmatrix} 1.5 & -0.5 & 0 & \dots & 0 \\ -0.5 & 1 & -0.5 & \dots & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & \dots & -0.5 & 1 & -0.5 \\ 0 & \dots & 0 & -0.5 & 1.5 \end{bmatrix}$$

which is positive definite. To see why, $\mathcal{H}$ can be decomposed into the sum of two matrices:

$$\mathcal{H} = \begin{bmatrix} 0.5 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & \dots & 0 & 0 & 0 \\ 0 & \dots & 0 & 0 & 0.5 \end{bmatrix} + \begin{bmatrix} 1 & -0.5 & 0 & \dots & 0 \\ -0.5 & 1 & -0.5 & \dots & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & \dots & -0.5 & 1 & -0.5 \\ 0 & \dots & 0 & -0.5 & 1 \end{bmatrix}$$

The first matrix is positive semidefinite (with two positive eigenvalues and all others zero), and the second matrix is positive definite because the determinant of the ℓ-th leading principal minor is given by $\left(\frac{1}{2}\right)^{\ell}(\ell+1)$ for $\ell = 1, ..., k$. Hence, first-order conditions are sufficient for optimality.

For uniqueness, say there were two optimal sets of points, $\mathcal{P}$ and $\mathcal{P}^*$. Based on the positive definite Hessian, a convex combination strictly decreases the objective function value. And if $\ell b \leq \mathcal{P}_1 \leq ... \leq \mathcal{P}_k \leq (\ell+1)b$ and $\ell b \leq \mathcal{P}_1^* \leq ... \leq \mathcal{P}_k^* \leq (\ell+1)b$ both hold, then their convex combination satisfies $\ell b \leq \gamma\mathcal{P}_1 + (1-\gamma)\mathcal{P}_1^* \leq ... \leq \gamma\mathcal{P}_k + (1-\gamma)\mathcal{P}_k^* \leq (\ell+1)b$. Hence, the convex combination is feasible. This contradicts optimality of the initial $\mathcal{P}$ and $\mathcal{P}^*$. □

A.3 Proof of Lemma 2

First I need to show that prices are a supermartingale for $p_t \in [i\varepsilon + \varepsilon/2, (i+1)\varepsilon]$ for $i \in \mathbb{Z}$. The first moment of LHS and RHS of the market maker's rule (6) yields

$$\mathbb{E}[p_{t+1}|p_t] - p_t = c\frac{\hat{\mathbb{E}}[p_{t+1}|p_t] - p_t}{\rho \mathrm{Var}(p_{t+1} - p_t|p_t) + \tau} \tag{15}$$

Note the hat in (15), as opposed to a tilde. I am first constructing equilibrium prices using $\hat{\mathbb{E}}[p_{t+1}|p_t]$, then showing (11). If (11) holds, then $\hat{\mathbb{E}}[p_{t+1}|p_t] = \tilde{\mathbb{E}}[p_{t+1}|p_t]$. Consequently, equilibrium prices do not change when $\tilde{\mathbb{E}}[p_{t+1}|p_t]$ is used in the demand function. Because I care only about signing (15), it suffices to analyze the numerator on the RHS. The mathematical

requirement for a price supermartingale becomes[18]

$$\text{mod}(-p, \varepsilon) - \varepsilon/2 \leq 0, \text{ for } p \in [i\varepsilon + \varepsilon/2, (i+1)\varepsilon]$$

Setting $i = 0$ without loss of generality because the modulo is periodic, this expression can be rewritten as

$$(-p + \varepsilon) - \varepsilon/2 \leq 0, \text{ for } p \in [\varepsilon/2, \varepsilon]$$

which is algebraically true. Symmetrically, I need to address the submartingale case. I exercise caution here because the lower bound of the price range $p_t \in [i\varepsilon, i\varepsilon + \varepsilon/2]$ overlaps with the adjacent (supermartingale) price range. While it is not the only way to do so, here I write the proof concretely for the case where non-unique argmins in (3) are always selected using the *smaller* posterior mean. This is consistent with the modulo formulation because $\text{mod}(-i\varepsilon, \varepsilon) - \varepsilon/2 = -\varepsilon/2$ for $i \in \mathbb{Z}$. Using periodicity and the same steps as above, I require that

$$(-p + \varepsilon) - \varepsilon/2 \geq 0, \text{ for } p \in (0, \varepsilon/2]$$

which is, again, algebraically true.

The second part of the argument is to show that $|\mathbb{E}[p_{t+1}|p_t] - p_t| \leq \varepsilon/2$. If this is true, then for $p_t \in [i\varepsilon + \varepsilon/2, (i+1)\varepsilon]$,

$$0 \leq p_t - \hat{\mathbb{E}}[p_{t+1}|p_t] \leq \varepsilon/2, \text{ and } -\varepsilon/2 \leq \mathbb{E}[p_{t+1}|p_t] - p_t \leq 0$$

where the first two inequalities follow from the fact that $\hat{\mathbb{E}}[p_{t+1}|p_t] = i\varepsilon + \varepsilon/2$ when $p_t \in [i\varepsilon + \varepsilon/2, (i+1)\varepsilon]$, and the second two inequalities follow from $|\mathbb{E}[p_{t+1}|p_t] - p_t| \leq \varepsilon/2$ along with the supermartingale argument. Add these two pairs of inequalities together to get the desired condition:

$$-\varepsilon/2 \leq \mathbb{E}[p_{t+1}|p_t] - \hat{\mathbb{E}}[p_{t+1}|p_t] \leq \varepsilon/2 \tag{16}$$

A similar argument applies for the case when $p_t \in (i\varepsilon, i\varepsilon + \varepsilon/2]$. Then

$$-\varepsilon/2 \leq p_t - \hat{\mathbb{E}}[p_{t+1}|p_t] \leq 0, \text{ and } 0 \leq \mathbb{E}[p_{t+1}|p_t] - p_t \leq \varepsilon/2$$

which can be added together to, again, get condition (16).

To show that $|\mathbb{E}[p_{t+1}|p_t] - p_t| \leq \varepsilon/2$, consider again condition (15) and take absolute values:

$$|\mathbb{E}[p_{t+1}|p_t] - p_t| = c\frac{|\hat{\mathbb{E}}[p_{t+1}|p_t] - p_t|}{\rho \text{Var}(p_{t+1} - p_t|p_t) + \tau} = c\frac{|\hat{\mathbb{E}}[p_{t+1}|p_t] - p_t|}{\rho c^2 \text{Var}(u_t) + \tau} \leq |\hat{\mathbb{E}}[p_{t+1}|p_t] - p_t| \leq \varepsilon/2$$

where the second equality follows from the variance condition (9), the first inequality follows

[18] To see that $\hat{\mathbb{E}}[p_{t+1}|p] - p$ can be written as $\text{mod}(-p, \varepsilon) - \varepsilon/2$, I verify three properties. First, it has slope -1 almost everywhere. Second, it takes on value zero when $p = i\varepsilon + \varepsilon/2$ for $i \in \mathbb{Z}$. Third, discontinuities occur at $p = i\varepsilon$ for $i \in \mathbb{Z}$.

from the fact that $c = \tau/2$ implies that $\frac{c}{\rho c^2 \mathrm{Var}(u_t)+\tau} \leq 1$, and the final inequality follows from the construction of $\hat{\mathbb{E}}[p_{t+1}|p_t]$. $\square$

A.4 Proof of Lemma 3

The conditional distribution of $\mathrm{mod}(p_{t+1}, \varepsilon)$ is

$$\begin{aligned}\Pr(\mathrm{mod}(p_{t+1}, \varepsilon) \leq y|p) =& \Pr(\mathrm{mod}(p + cx(p) + cu, \varepsilon) \leq y) \\ =& \Pr(\mathrm{mod}(\mathrm{mod}(p, \varepsilon) + \mathrm{mod}(cx(p), \varepsilon) + \mathrm{mod}(cu, \varepsilon), \varepsilon) \leq y) \\ =& \Pr(\mathrm{mod}(\mathrm{mod}(p, \varepsilon) + \mathrm{mod}(cx(\mathrm{mod}(p, \varepsilon)), \varepsilon) + \mathrm{mod}(cu, \varepsilon), \varepsilon) \leq y)\end{aligned}$$

where the second equality follows from properties of the modulo function and the third equality follows from the fact that demand $x(p)$ is periodic. Calling $\mathcal{P}_{t+1} \equiv \mathrm{mod}(p_{t+1}, \varepsilon)$,

$$\Pr(\mathcal{P}_{t+1} \leq y|\mathcal{P}) = \Pr(\mathrm{mod}(\mathcal{P} + \mathrm{mod}(cx(\mathcal{P}), \varepsilon) + \mathrm{mod}(cu, \varepsilon), \varepsilon) \leq y)$$

Call this distribution $F^{mod}(y|\mathcal{P})$. It is absolutely continuous because

$$\begin{aligned}&\Pr(\mathrm{mod}(\mathcal{P} + \mathrm{mod}(cx(\mathcal{P}), \varepsilon) + \mathrm{mod}(cu, \varepsilon), \varepsilon) \leq y) \\ =& \sum_{i=1}^{3} \Pr(\mathcal{P} + \mathrm{mod}(cx(\mathcal{P}), \varepsilon) + \mathrm{mod}(cu, \varepsilon) - [i-1]\varepsilon \leq y) \\ -& \sum_{i=1}^{3} \Pr(\mathcal{P} + \mathrm{mod}(cx(\mathcal{P}), \varepsilon) + \mathrm{mod}(cu, \varepsilon) - [i-1]\varepsilon \leq 0)\end{aligned}$$

To show that $\mathrm{mod}(cu, \varepsilon)$ in the resulting term has a density, which I call $\tilde{h}(y)$, I claim that it is, explicitly,

$$\tilde{h}(y) \equiv \sum_{j \in \mathbb{Z}} h([y + j\varepsilon]/c)/c$$

To show that $\tilde{h}(y)$ is, indeed, the density of $\mathrm{mod}(cu, \varepsilon)$, I calculate its distribution

$$\begin{aligned}\Pr(\mathrm{mod}(cu, \varepsilon) \leq y) =& \sum_{j \in \mathbb{Z}} [\Pr(cu - j\varepsilon \leq y) - \Pr(cu - j\varepsilon \leq 0)] \\ =& \sum_{j \in \mathbb{Z}} [\Pr(u \leq [y + j\varepsilon]/c) - \Pr(u \leq j\varepsilon/c)]\end{aligned}$$

and, as required,

$$\int_0^y \tilde{h}(x)dx = \int_0^y \lim_{N\to\infty} \sum_{j=-N}^{N} h([x+j\varepsilon]/c)\frac{dx}{c} = \lim_{N\to\infty} \sum_{j=-N}^{N} \int_0^y h([x+j\varepsilon]/c)\frac{dx}{c}$$
$$=\Pr(\text{mod}(cu,\varepsilon) \leq y)$$

where the second equality follows from the monotone convergence theorem. Back to the original line of reasoning, $f^{mod}(y|\mathcal{P})$ can then be written as

$$f^{mod}(y|\mathcal{P}) = \sum_{i=1}^{3} \tilde{h}(y + [i-1]\varepsilon - \mathcal{P} - \text{mod}(cx(\mathcal{P}),\varepsilon))$$

I conclude that a conditional price density, $f^{mod}(y|\mathcal{P})$, exists.

I consider (12) as a fixed point of a linear operator $\mathcal{T}(g^{mod})$, mapping from the space of symmetric, non-negative functions that integrate to one on $[0,\varepsilon]$ to itself. Call this space K. The goal is to apply the Schauder fixed point theorem to this linear operator. $\mathcal{T}(g^{mod})$ integrates to one on $[0,\varepsilon]$ because

$$\int_0^\varepsilon \mathcal{T}(g^{mod})(x)dx = \int_0^\varepsilon \int_0^\varepsilon f^{mod}(x|\mathcal{P})g^{mod}(\mathcal{P})d\mathcal{P}dx = \int_0^\varepsilon \int_0^\varepsilon f^{mod}(x|\mathcal{P})dx g^{mod}(\mathcal{P})d\mathcal{P}$$
$$= \int_0^\varepsilon g^{mod}(\mathcal{P})d\mathcal{P}$$
$$=1$$

where the second equality follows from Tonelli's Theorem and the last equality follows from a domain with integral one. To see that $\mathcal{T}(g^{mod})(x)$ is symmetric, evaluate at $\varepsilon - x$:

$$\begin{aligned}
\mathcal{T}(g^{mod})(\varepsilon - x) &= \int_0^\varepsilon f^{mod}(\varepsilon - x|\mathcal{P})g^{mod}(\mathcal{P})d\mathcal{P} \\
&= \int_0^\varepsilon f^{mod}(x|\varepsilon - \mathcal{P})g^{mod}(\mathcal{P})d\mathcal{P} \\
&= \int_0^\varepsilon f^{mod}(x|r)g^{mod}(\varepsilon - r)dr \\
&= \int_0^\varepsilon f^{mod}(x|r)g^{mod}(r)dr \\
&= \mathcal{T}(g^{mod})(x)
\end{aligned}$$

where the second equality is proven below, the third equality follows from the change of variable

$r = \varepsilon - \mathcal{P}$, and the fourth equality follows from a symmetric domain. Below, I show that

$$\begin{aligned}
\Pr(\mathcal{P}_{t+1} \leq \varepsilon - y|\mathcal{P}) =& \Pr(\text{mod}(\mathcal{P} + \text{mod}(cx(\mathcal{P}), \varepsilon) + \text{mod}(cu, \varepsilon), \varepsilon) \leq \varepsilon - y) \\
=& \Pr(-\text{mod}(\mathcal{P} + \text{mod}(cx(\mathcal{P}), \varepsilon) + \text{mod}(cu, \varepsilon), \varepsilon) \geq y - \varepsilon) \\
=& \Pr(\varepsilon - \text{mod}(p + cx(\mathcal{P}) + cu, \varepsilon) \geq y) \\
=& \Pr(\text{mod}(\varepsilon - p - cx(\mathcal{P}) - cu, \varepsilon) \geq y) \\
=& \Pr(\text{mod}(\varepsilon - p + cx(\varepsilon - \mathcal{P}) + cu, \varepsilon) \geq y) \\
=& \Pr(\text{mod}(\varepsilon - \mathcal{P} + \text{mod}(cx(\varepsilon - \mathcal{P}), \varepsilon) + \text{mod}(cu, \varepsilon), \varepsilon) \geq y) \\
=& 1 - Pr(\mathcal{P}_{t+1} \leq y|\varepsilon - \mathcal{P})
\end{aligned}$$

for $0 < \mathcal{P} < \varepsilon$, and equalities are algebraic other than using $x(\varepsilon - \mathcal{P}) = -x(\mathcal{P})$, where $x(p)$ denotes demand, and noise trading symmetry in the fifth equality. Taking derivatives with respect to y yields the equality

$$f^{mod}(\varepsilon - y|\mathcal{P}) = f^{mod}(y|\varepsilon - \mathcal{P}),$$

as required.

The Schauder fixed point theorem requires that the space, K, be convex, closed, and nonempty. These properties can be explicitly verified. The theorem also requires that the operator be continuous, which is equivalent to bounded when it is linear. But this was just shown above: $||\mathcal{T}(g^{mod})||_1 = ||g^{mod}||_1$ because both $\mathcal{T}(g^{mod})$ and g^{mod} integrate to one. Finally, the theorem requires that the image of $\mathcal{T}$ be contained in a compact subset of K. By Prokhorov's theorem, a collection of probability measures is tight if and only if its closure is compact (with the topology of weak convergence). But because K consists of densities with bounded support, tightness is immediate. □

A.5 Proof of Theorem 2

First I show that the conditional price distribution $f(y|p)$ is conditionally symmetric. This does not follow from Lemma 3, which only showed this property for $f^{mod}(y|\mathcal{P})$. Consider some $p = \varepsilon/2 - \delta$ for $0 \leq \delta < \varepsilon/2$. Evaluating the conditional price distribution at such a p,

$$\begin{aligned}
F(y|\varepsilon/2 - \delta) = Pr(\varepsilon/2 - \delta + cx(\varepsilon/2 - \delta) + cu \leq y) =& Pr(\varepsilon/2 - \delta - cx(\varepsilon/2 + \delta) - cu \leq y) \\
=& Pr(\varepsilon/2 + \delta + cx(\varepsilon/2 + \delta) + cu \geq \varepsilon - y) \\
=& 1 - F(\varepsilon - y|\varepsilon/2 + \delta)
\end{aligned}$$

where the second equality importantly follows from i) the fact that the modulo demand satisfies $x(\varepsilon/2 - \delta) = -x(\varepsilon/2 + \delta)$ and ii) symmetry of noise trading. The third equality is algebraic. Taking a derivative of both sides with respect to y, $f(y|\varepsilon/2 - \delta) = f(\varepsilon - y|\varepsilon/2 + \delta)$ for all y and $0 \leq \delta < \varepsilon/2$.

Next I must show that[19]

$$\text{Cov}^{mod}(\mathbb{E}[p_{t+1} - p|p], p) = \int_0^{\varepsilon/2} \mathbb{E}[p_{t+1} - p|p] p g^{mod}(p) dp + \int_{\varepsilon/2}^{\varepsilon} \mathbb{E}[p_{t+1} - p|p] p g^{mod}(p) dp < 0$$

With the change of variable $x = \varepsilon - p$, this becomes

$$-\int_{\varepsilon}^{\varepsilon/2} \mathbb{E}[p_{t+1} + x - \varepsilon|\varepsilon - x](\varepsilon - x) g^{mod}(\varepsilon - x) dx + \int_{\varepsilon/2}^{\varepsilon} \mathbb{E}[p_{t+1} - p|p] p g^{mod}(p) dp < 0$$

Invoking symmetry of $g^{mod}(p)$ from Lemma 3,

$$\int_{\varepsilon/2}^{\varepsilon} \mathbb{E}[p_{t+1} + x - \varepsilon|\varepsilon - x](\varepsilon - x) g^{mod}(x) dx + \int_{\varepsilon/2}^{\varepsilon} \mathbb{E}[p_{t+1} - p|p] p g^{mod}(p) dp < 0$$

Focusing on the term inside of the first integral,

$$\begin{aligned}
\mathbb{E}[p_{t+1} + x - \varepsilon|\varepsilon - x] &= \int_{-\infty}^{\infty} (y + x - \varepsilon) f(y|\varepsilon - x) dy \\
&= \int_{-\infty}^{\infty} (y + x - \varepsilon) f(\varepsilon - y|x) dy \\
&= \int_{-\infty}^{\infty} (-z + x) f(z|x) dz
\end{aligned}$$

which uses conditional symmetry of $f(y|p)$ and a change of variable $z = \varepsilon - y$. Altogether, my desired inequality is

$$\begin{aligned}
&-\int_{\varepsilon/2}^{\varepsilon} \mathbb{E}[p_{t+1} - x|x](\varepsilon - x) g^{mod}(x) dx + \int_{\varepsilon/2}^{\varepsilon} \mathbb{E}[p_{t+1} - p|p] p g^{mod}(p) dp \\
&= \int_{\varepsilon/2}^{\varepsilon} \mathbb{E}[p_{t+1} - p|p](2p - \varepsilon) g^{mod}(p) dp < 0
\end{aligned}$$

which is true because $2p - \varepsilon > 0$ for all $\varepsilon/2 < p \leq \varepsilon$, and $\mathbb{E}[p_{t+1} - p|p] < 0$ again for all $\varepsilon/2 < p \leq \varepsilon$. This latter point follows from the supermartingale argument detailed in Lemma 2. Now say that $g^{mod}(p)$ puts zero mass on prices satisfying $\varepsilon/2 < p \leq \varepsilon$. By symmetry, it must only put positive mass on the point $p = \varepsilon/2$, a contradiction to absolute continuity. Based on this argument, $g^{mod}(p)$ must put some positive mass on prices satisfying $\varepsilon/2 < p \leq \varepsilon$, completing the proof. □

[19] Note that, using the same steps as those shown in this corollary,

$$\int_0^{\varepsilon/2} \mathbb{E}[p_{t+1} - p|p] g^{mod}(p) dp + \int_{\varepsilon/2}^{\varepsilon} \mathbb{E}[p_{t+1} - p|p] g^{mod}(p) dp = 0$$

which is why the covariance can be written in this way.

A.6 Proof of Corollary 1

Note that the sub- and super-martingale restrictions on $\mathbb{E}[p_{t+1}|p_t]$ have already been shown in Lemma 2. Inequalities there are strict when $p \neq \varepsilon/2$. For restrictions on $\overline{\mathbb{E}}[p_{t+1}|p_t]$, I take an expectation of both LHS and RHS of the market maker rule (6) including arbitrageurs,

$$\overline{\mathbb{E}}[p_{t+1}|p_t] - p_t = c\frac{\tilde{\mathbb{E}}[p_{t+1}|p_t] - p_t]}{\rho \mathrm{Var}(p_{t+1} - p_t|p_t) + \tau} + c\frac{\overline{\mathbb{E}}[p_{t+1}|p_t] - p_t}{\hat{\rho}\mathrm{Var}(p_{t+1} - p_t|p_t) + \tau}$$

where $\hat{\rho}$ denotes risk aversion of the rational traders. Note that the addition of rational traders does not affect the price variance. Solving for the true mean of returns and plugging in $\mathrm{Var}(p_{t+1} - p_t|p_t) = c^2\mathrm{Var}(u_t)$,

$$\overline{\mathbb{E}}[p_{t+1}|p_t] - p_t\left(1 - \frac{c}{\hat{\rho}c^2\mathrm{Var}(u_t) + \tau}\right) = c\frac{\tilde{\mathbb{E}}[p_{t+1}|p_t] - p_t]}{\rho c^2\mathrm{Var}(u_t) + \tau} \tag{17}$$

First, I must verify that

$$1 - \frac{c}{\hat{\rho}c^2\mathrm{Var}(u_t) + \tau} > 0$$

so that price expectations inherit the sign of coarse Bayesian trades. But this inequality holds when $c = \tau/2$. Second, the additional trading by arbitrageurs might potentially cause Lemma 2 to fail. However, I can show that this is not the case because, dividing (17) by $1 - \frac{c}{\hat{\rho}c^2\mathrm{Var}(u_t)+\tau}$, the term that multiplies the coarse expectation in (17) is

$$\begin{aligned}\frac{c/(\rho c^2\mathrm{Var}(u_t) + \tau)}{1 - c/(\hat{\rho}c^2\mathrm{Var}(u_t) + \tau)} \leq 1 &\iff c/(\rho c^2\mathrm{Var}(u_t) + \tau) + c/(\hat{\rho}c^2\mathrm{Var}(u_t) + \tau) \leq 1 \\ &\Longleftarrow \frac{2c}{\tau} = 1\end{aligned}$$

But this last equality holds when $c = \tau/2$. If Lemma 2 holds, then coarse expectations are calculated in the same way with and without rational traders. Without rational traders, (6) implies that

$$\mathbb{E}[p_{t+1}|p_t] - p_t = c\frac{\tilde{\mathbb{E}}[p_{t+1}|p_t] - p_t]}{\rho c^2\mathrm{Var}(u_t) + \tau} \tag{18}$$

Then comparing (17) to (18), I require that

$$c\frac{\tilde{\mathbb{E}}[p_{t+1}|p_t] - p_t]}{\rho c^2\mathrm{Var}(u_t) + \tau}\Bigg/\left(1 - \frac{c}{\hat{\rho}c^2\mathrm{Var}(u_t) + \tau}\right) > c\frac{\tilde{\mathbb{E}}[p_{t+1}|p_t] - p_t]}{\rho c^2\mathrm{Var}(u_t) + \tau} \text{ for } p_t \in (0, \varepsilon/2)$$

which simplifies to

$$1 - \frac{c}{\hat{\rho}c^2\mathrm{Var}(u_t) + \tau} < 1$$

because the coarse Bayesian buys in this price range by Lemma 2. This last inequality is true because all parameter values in the fraction are positive. Similarly, I require that

$$c\frac{\tilde{\mathbb{E}}[p_{t+1}|p_t]-p_t]}{\rho c^2\text{Var}(u_t)+\tau}\Bigg/\left(1-\frac{c}{\hat{\rho}c^2\text{Var}(u_t)+\tau}\right)<c\frac{\tilde{\mathbb{E}}[p_{t+1}|p_t]-p_t]}{\rho c^2\text{Var}(u_t)+\tau}\text{ for } p_t\in(\varepsilon/2,\varepsilon)$$

which again simplifies to

$$1-\frac{c}{\hat{\rho}c^2\text{Var}(u_t)+\tau}<1$$

because the coarse Bayesian sells in this price range by Lemma 2. □

B Zero Profits

In this section, I explore the consequence of increasing the mass of coarse Bayesian traders on the price elasticity, c. Call this mass $M\geq 1$, where the baseline case is captured by $M=1$. Note that, because transaction costs are quadratic, increasing the mass of traders is *not* equivalent to increasing the trade per individual. I claim that $\frac{\tau}{2M}\leq c\leq\frac{\tau}{2}$ or, in words, the price elasticity must decrease. To see why, first say that $c<\frac{\tau}{2M}$. Then, explicitly writing out the market maker's expected profits,

$$\begin{aligned}
&\int_0^\varepsilon \mathbb{E}\left[\frac{\tau}{2}u_t^2+\frac{\tau}{2}Mx(p)^2-c\left(u_t+Mx(p)\right)^2\right]g^{mod}(p)dp\\
>&\int_0^\varepsilon \mathbb{E}\left[\frac{\tau}{2}u_t^2+\frac{\tau}{2}Mx(p)^2-\frac{\tau}{2M}\left(u_t+Mx(p)\right)^2\right]g^{mod}(p)dp\\
=&\int_0^\varepsilon \mathbb{E}\left[\frac{\tau}{2}u_t^2-\frac{\tau}{2M}u_t^2\right]g^{mod}(p)dp\\
\geq&0
\end{aligned}$$

where the line of reasoning above is purely algebraic and relies on $\text{Var}(u_t)>0$ and $\mathbb{E}[u_t]=0$. Note that $g^{mod}(p)$ solves (12). In words, profits will be positive when the price elasticity is too small. Next, say that $c>\tau/2$. Then,

$$\begin{aligned}
&\int_0^\varepsilon \mathbb{E}\left[\frac{\tau}{2}u_t^2+\frac{\tau}{2}Mx(p)^2-c\left(u_t+Mx(p)\right)^2\right]g^{mod}(p)dp\\
<&\int_0^\varepsilon \mathbb{E}\left[\frac{\tau}{2}u_t^2+\frac{\tau}{2}Mx(p)^2-\frac{\tau}{2}\left(u_t+Mx(p)\right)^2\right]g^{mod}(p)dp\\
=&\int_0^\varepsilon \mathbb{E}\left[\frac{\tau}{2}Mx(p)^2-\frac{\tau}{2}M^2x(p)^2\right]g^{mod}(p)dp\\
\leq&0
\end{aligned}$$

where the line of reasoning above is, again, purely algebraic and relies on $\mathrm{Var}(u_t) > 0$ and $\mathbb{E}[u_t] = 0$. In words, profits will be negative when the price elasticity is too large. While $\frac{\tau}{2M} \leq c \leq \frac{\tau}{2}$ are necessary conditions for zero profits, sufficient conditions are more involved because both the ergodic density $g^{mod}(p)$ and demand $x(p)$ depend on c.

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